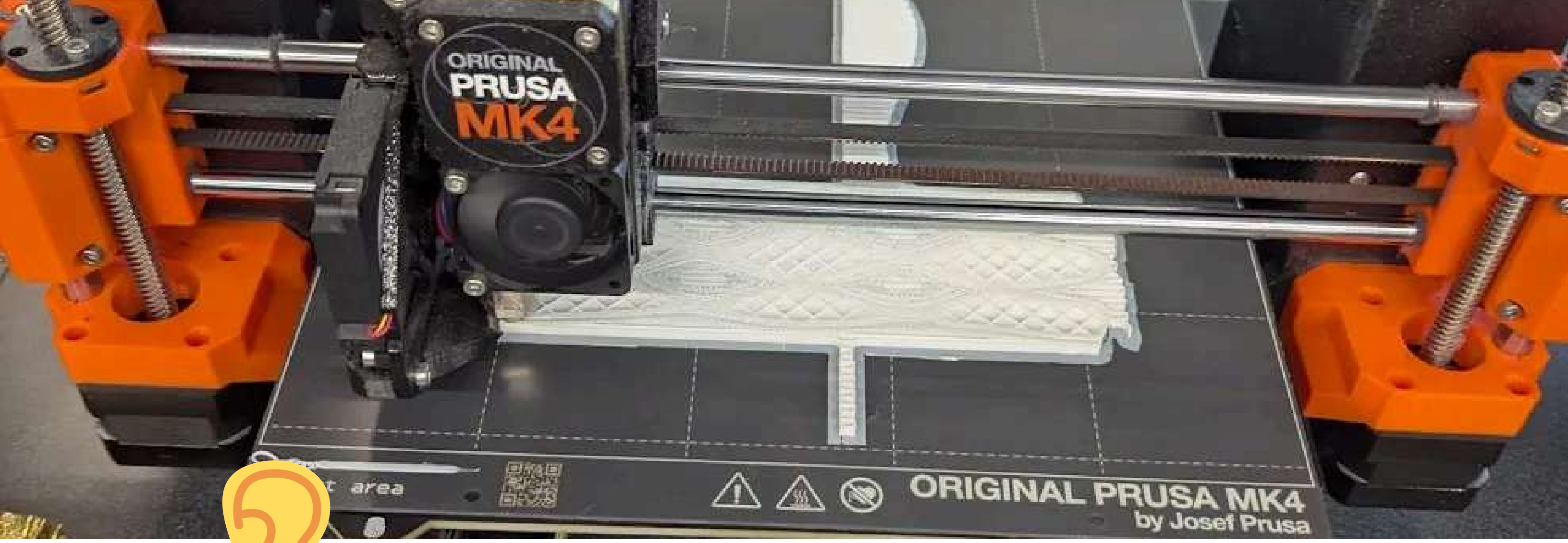


Design Project 4: Earring application gun

Overview

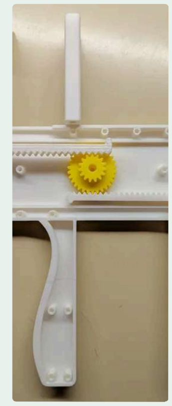
🔧 Skills Applied/Learned

- ✅ CAD with more complex geometry, freeform modelling
- ✅ Assembly: making and constraining gears + rack and pinion
- ✅ 3D Printing
- ✅ Interpersonal skills
- ✅ Time Management
- ✅ Collaboration & Coordination: Splitting work
- ✅ Communication: Oral and Written
- ✅ Presentation Skills

🎯 Patient Summary

- We were introduced to **real patients** and challenged to identify and try to improve one of their challenges.
- **Jany** has MS, and mentioned having trouble putting on stud earrings because of limited dexterity, so she now wears hoops instead
    - We hoped to create device that allowed her to independently put studs on again

Printed Models of Final Design



📕 **Background Info, if needed:**

- **Stud Earrings**: Smaller, harder to handle
- **MS**: Multiple Sclerosis, a chronic autoimmune disease where the immune system attacks myelin sheaths of nerves. Symptoms include fatigue, numbness, tingling, walking difficulty, and **coordination** issues.

Design Process

🧰 **Identifying Goals: the problem to solve**

🎯 Objectives

- Should be **comfortable** to use
- Should be **efficient** to use
- Should be **portable** and not bulky
- Should improve **independence**

🔗 Constraints

- adhere to health and safety **standards**
- Must be usable with Jany’s current **finger dexterity**
- Must be **safe** to use
- Must not **worsen her condition**

💡 **Initial Thoughts**

- **Scissor** mechanism
  - simple
  - adaptable to multiple functions
    - potentially nail clippers too
- **Motorized closing possibilities:**
  - wind up string
  - cam and linkage
  - linear actuator or pushing it down
- **Materials**
  - Plastic: cheap and usable

🧠 **Brainstorm and discussion with group**

We decided to **focus** on the earing applying function first

We divided the work, someone else will do the earing holder and backing holder

I will CAD the scissor mechanism in a **basic** form

We need further thought one the specifics for loading the earing pieces, how can it be **easier** than just using **hands**?

Lowest Fidelity Prototype/model from discussion

Prototyping

I created the CAD models for the "gun" portion of the design.

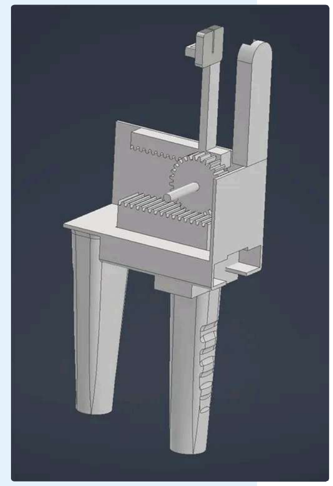
1 The **initial idea** was to use a scissor mechanism.

We liked how slim it was at first.

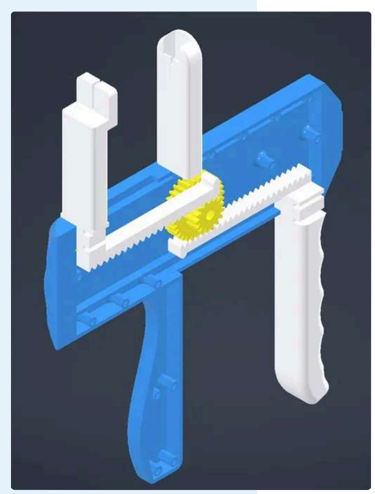
However, it would have to be much bigger, too big, to create **good alignment** and the required **leverage.** We eventually decided to abandon the scissor idea.

Before finishing we also abandoned the **multi tool idea**, without the scissor, we would focus on a mechanism **solely suited** to the earring.

Initial scissor mechanism



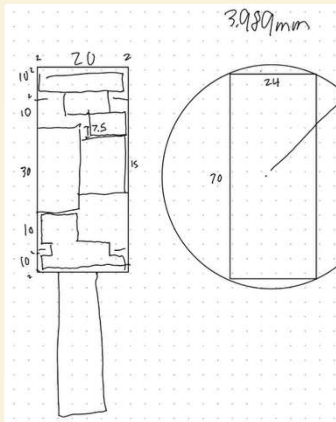
Second bare bones rack and pinion model



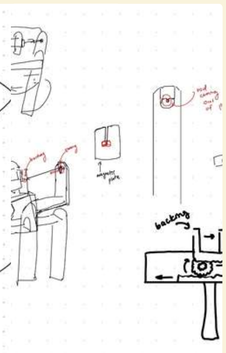
Most finished model

➡️ Starting on a new prototype, we decided to move away from the **scissor** mechanism.

I began to work on measurements and sketches for a **rack and pinion/gun design**



Cross section



full body + team members earing holder sketch

⏭️ Final changes

Adjusted the locations of a few of the pegs and shifted the gaps for the handle and holders to slide because my measurements were slightly off

Created an assembly and constrained the parts to test virtual

Then we began printing

New CAD Model

purposely **bare bones**, very **unaesthetically** pleasing and not designed for assembly

An assembly file with proper **constraints** showed that the concept would work.

We printed an made a mock up to test it **physically** too, as seen in the right.

✅ The final Product

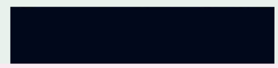
Conclusion + Reflection

Features Diagram

🏗️ My Overall Contributions

- Created CAD model of the **body, handle and gear mechanism**
- Created **final assembly** of CAD model
- Coordinated **3D printing** of the model
- Contributes slides to the slideshow for presentation
- Appeared and create few scenes in our video
- Helped **physically** assemble the final prototype

motion, bringing the backing



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🎊 What Went Well (expand for details)

▼ Communication

Our team communicated **well**. We focused on making a solely mechanical project this time which made it

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🪞 Challenges and Lessons

▼ Scope

While deciding on our initial idea, we had trouble deciding between **multiple** good ideas to determine which one fit best into the time frame we had. We got a better idea of